

Week 2 — Tuesday Lab: Search, Filters & Visuals

IM102 – Advanced Database Systems | 3 Hours | Computer Lab

Today's Goal

Make your product list interactive — search, filter by category, highlight important info.

Example: Dynamic WHERE Clause

Instead of writing separate queries for every filter, build the query piece by piece:

```
$sql = "SELECT ... FROM products p JOIN categories c ... WHERE 1=1";

if (!empty($search)) {
    $sql .= " AND p.name LIKE '%" . $conn->real_escape_string($search) .
    "%'";
}
if (!empty($category)) {
    $sql .= " AND c.name = '" . $conn->real_escape_string($category) . "'";
}
```

WHERE 1=1 is just a trick so you can always append AND ... without worrying about whether it's the first condition.

Example: Database-Powered Dropdown

Instead of hardcoding category names in HTML, query them:

```
$categories = $conn->query("SELECT DISTINCT name FROM categories ORDER BY
name");
```

```
<select name="category">
  <option value="">All Categories</option>
  <?php while ($c = $categories->fetch_assoc()): ?>
```

```
<option><?= $c['name'] ?></option>
<?php endwhile; ?>
</select>
```

Now if you add a new category in the database, it automatically appears in every filter dropdown. No HTML changes needed.

Example: Aggregate Stats

```
$stats = $conn->query("
    SELECT COUNT(*) AS total, SUM(stock) AS total_stock,
           SUM(price * stock) AS total_value
    FROM products
")->fetch_assoc();
```

Your Turn

Add a search bar that filters products by name or description. Add a category dropdown that filters by category. Make them work together — search "laptop" in "Electronics" should show only electronics matching "laptop".

Highlight products with stock below 20 in red. Add summary cards showing total products, total stock, total inventory value, and count of low-stock items.

The numbers in your cards should change as you filter — that's how you know the queries are right.